

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276622

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Svitavsky; Tomas et al.

SEAT ASSEMBLY

Abstract

A track assembly comprising a first track having a curved configuration; a second track slidably connected to the first track; a seat assembly connected to move with the second track, the seat assembly comprising a seat base and a seat back adjustable to a plurality of seat positions relative to the second track; a lock configured to selectively lock the first track and the second track; and a release lever configured to automatically actuate the lock.

Inventors: Svitavsky; Tomas (Pilsen, CZ), Bilek; Tomas (Pilsen, CZ), Novakova; Michaela (Pilsen, CZ), El Alaoui; Abdessalam (Rabat, MA), Frank; Thomas (Ismaning, DE)

Applicant: Lear Corporation (Southfield, MI)

Family ID: 96738285

Assignee: Lear Corporation (Southfield, MI)

Appl. No.: 19/057368

Filed: February 19, 2025

Foreign Application Priority Data

DE 10 2024 106 186.3

Mar. 04, 2024

Publication Classification

Int. Cl.: B60N2/427 (20060101); B60N2/07 (20060101); B60N2/90 (20180101)

U.S. Cl.:

CPC B60N2/42745 (20130101); B60N2/074 (20130101); B60N2/0745 (20130101); B60N2002/924 (20180201)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This applications claims priority to and the benefit of German Patent Application 10 2024 106 186.3, filed Mar. 4, 2024, the disclosure of which is hereby incorporated by reference in its entirety as though fully set forth herein.

TECHNICAL FIELD

[0002] The present disclosure generally relates to seat assemblies, including seat assemblies that can, for example, be utilized in connection with vehicles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] While the claims are not limited to a specific illustration, an appreciation of various aspects may be gained through a discussion of various examples. The drawings are not necessarily to scale, and certain features may be exaggerated or hidden to better illustrate and explain an innovative aspect of an example. Further, the exemplary illustrations described herein are not exhaustive or otherwise limiting, and embodiments are not restricted to the precise form and configuration shown in the drawings or disclosed in the following detailed description. Exemplary illustrations are described in detail by referring to the drawings as follows:

[0004] FIG. 1 is a schematic view generally illustrating an embodiment of a seat assembly with a second track in a first position according to teachings of the present disclosure.

[0005] FIG. 2 is a schematic view generally illustrating an embodiment of a seat assembly with a second track in a second position according to teachings of the present disclosure.

[0006] FIG. 3 is a schematic view generally illustrating an embodiment of a seat assembly with a second track in a second position according to teachings of the present disclosure.

[0007] FIG. 4 is a schematic view generally illustrating an embodiment of a seat assembly in a first orientation with a second track in a first position according to teachings of the present disclosure.

[0008] FIG. 5 is a schematic view generally illustrating an embodiment of a seat assembly in a second orientation with a second track in a first position according to teachings of the present disclosure.

[0009] FIG. 6 is a schematic view generally illustrating an embodiment of a seat assembly in a second orientation with a second track in a second position according to teachings of the present disclosure.

[0010] FIG. 7 is a flow diagram generally illustrating an embodiment of a method of operating a seat assembly according to teachings of the present disclosure.

DETAILED DESCRIPTION

[0011] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described embodiments.

However, it will be apparent to one of ordinary skill in the art that the various described embodiments may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

[0012] Seat assemblies can be utilized in connection with vehicles that may experience crashes. Limiting the forces applied to occupants of the seat assemblies and/or limiting movement of the occupants during crashes is desirable. In some instances, the seat assemblies are not initially positioned in ideal positions for crashes, so automatically moving seat assemblies to positions that

better limit forces applied to occupants and/or better limit movement of occupants is desirable.

[0013] Referring to FIG. 1, a track assembly **20** includes a seat assembly **24** connected to a mounting surface **26** via a first track **60** and a second track **62**. An occupant **28** is shown seated on the seat assembly **24**. The seat assembly **24** includes a seat base **40** and a seat back **42**. At least one of the seat base **40** or the seat back **42** is rotatable relative to the other. For example, the seat back **42** is rotatable relative to the seat base **40** about a seat back axis **48** (e.g., a recliner axis) to provide various reclined positions for the occupant **28**, such as via a recliner **44**. The seat back axis **48** is parallel with the mounting surface **26** and perpendicular to the direction the seat assembly **24** is facing. The seat assembly **24** includes a head restraint **46** connected to a top of the seat back **42**. The track assembly **20** is shown in a vehicle **30**, but can be utilized in other applications.

[0014] The second track **62** slidably connected to the first track **60**. The first track **60** includes a curved configuration with a radius **R1** and is mounted to the mounting surface **26**. The first track **60** is parallel with an X-Z plane and is curved such that a distance in the Z-direction of the first track **60** from the mounting surface **26** varies along the X-direction. The first track **60** includes a middle section **70** that is disposed closest to the mounting surface **26**, a front section **72** that curves forward and upward from the middle section **70**, and a rear section **74** that curves rearward and upward from the middle section **70**. An angular extent **76** of the first track **60** can, for example, be at least 30 degrees and less than or equal to 180 degrees, at least 30 degrees and less than or equal to 120 degrees, at least 30 degrees and less than or equal to 90 degrees, or at least 40 degrees and less than or equal to 50 degrees, among others. The angular extent **78** of the second track **62** is less than the angular extent **76** of the first track **60**, which can include the angular extent **78** being at least 50% and less than or equal to 90% of the angular extent **76**. For example, the angular extent **78** can be at least 25 degrees and less than or equal to 40 degrees. The first track **60** is sufficiently longer than the second track **62** to allow sliding movement of at least 10 degrees, and, at least with some configurations, at least 30 degrees. The extent of sliding movement of the second track **62** is less than the angular extent **76** of the first track **60**, such as by at least 10 degrees, and can be determined via positioning of one or more end stops **150**, **152**. The radius **R1** of the first track **60** can, for example, be at least 300 mm (11.8 in) and less than or equal to 900 mm (35.4 in), or less than 300 mm or greater than 900 mm. With some examples, the radius **R1** can be 600 mm, which can result in 10 degrees of angular movement of the second track **62** for every 100 mm of circumferential displacement along the first track **60** (e.g., 200 mm of circumferential displacement would result in 20 degrees of angular movement).

[0015] A center **66** of a circle **64** partially defined by the first track **60** and having the radius **R1** defines a rotational axis **80** about which the second track **62** and the seat assembly **24** connected thereto rotate. The seat base **40** and the seat back **42** are positioned such that an expected center of gravity **68** of the occupant **28** is disposed above the second track **62** and below (e.g., closer to the mounting surface **26**) the center **66** of the circle **64** and the rotational axis **80**. Such a configuration, with the center of gravity **68** offset vertically below the center **66** and the rotational axis **80**, tends to cause the seat assembly **24** to rotate about the rotational axis **80** during a crash.

[0016] The track assembly **20** includes a lock **90** that is configured to selectively lock the first track **60** and the second track **62**. The lock **90** is connected to the second track **62** to selectively engage the first track **60** to selectively allow and prevent sliding movement of the second track **62** along the first track **60**. The lock **90** can, for example, include one or more pins **92** movable in the radial direction to selectively engage corresponding apertures **94** of the first track **60**. The apertures **94** are spaced along the first track **60**, such as to provide various rotational seat positions.

[0017] The track assembly **20** includes a release lever **100** configured to automatically unlock the lock **90**. The release lever **100** is operably coupled to the lock **90** such that operation of the release lever **100** unlocks the lock **90**, unlocking the second track **62** from the first track **60** to allow sliding movement of the second track **62**, or locks the lock **90**, locking the second track **62** with the first track **60** to prevent sliding movement of the second track **62**.

[0018] The track assembly **20** includes an actuator **110** operably coupled with the release lever **100**, such as to automatically actuate (e.g., lock, unlock, or both) the lock **90**. The actuator **110** includes a mechanical actuator **112**, a powered actuator **114**, or both. The mechanical actuator **112** can include, for example, a mass configured to move when certain forces are applied to the mass, such as crash forces during a vehicle crash. For example, in the event of a crash, the crash forces cause movement of the mass, which actuates the release lever **100** to unlock the lock **90**. The mechanical actuator **112** may operate without an external power source (e.g., electrical power, fluid power, etc.). The powered actuator **114** can include, for example, at least one of a motor, an electromagnet, a piezoelectric material, a cylinder, or a pyrotechnic device. The powered actuator **114** can additionally include a mechanical actuator.

[0019] The track assembly **20** includes a controller **130** and a crash sensor **132**. The controller **130** is in communication with the powered actuator **114**, the crash sensor **132**, and/or a track motor **240**. The controller **130** controls, at least in part, operation of the powered actuator **114**, such as in response to receiving information from the crash sensor **132** that a crash has occurred or is imminent. For example, when the crash sensor **132** senses that a crash has occurred or is imminent, the controller **130** controls the powered actuator **114** to operate the release lever **100** to unlock the lock **90** and allow crash forces and/or momentum to cause the second track **62** to move along the first track **60** from a second position (e.g., a reclined position), such as shown in FIG. 2, to a first position, such as shown in FIG. 1. Additionally or alternatively, the controller **130** is configured to operate the powered actuator **114** to operate the release lever **100** to lock the lock **90** when the second track **62** reaches the first position. The crash forces and/or momentum may allow for a greater rate of position adjustment than is feasible via an actuator (e.g., the track motor **240**) coupled to move the second track **62** along the first track **60**. For example, a motor powerful enough to move the second track **62** at a rate comparable to the crash forces/momentum may be too large to fit the available space, may be heavier than desired, and/or may consume more electrical power than available or desired. Additionally or alternatively, utilizing at least some of the crash forces/momentum to move the second track **62** may reduce the forces applied to the occupant **28**. The crash sensor **132** can include one or more of a variety of configurations, such as an accelerometer, radar, lidar, others, or combinations thereof.

[0020] The track assembly **20** optionally includes a track position sensor **134** configured to sense a position of the second track **62** relative to the first track **60**. The controller **130** can utilize information from the track position sensor **134** to determine whether to actuate the powered actuator **114** to actuate the release lever **100** to lock or unlock the lock **90**. For example, in accordance with the track position sensor **134** indicating that the second track **62** is not in the first position (e.g., the crash position) and the crash sensor **132** indicates a rear crash, the controller **130** operates the powered actuator **114** to actuate the release lever **100** to unlock the lock **90** and allows the second track **62** to move toward or to the first position, such as via the crash forces. Additionally or alternatively, in accordance with the track position sensor **134** indicating that the second track **62** is already in the first position with a rear crash, the controller **130** actuates the powered actuator **114** to actuate the release lever **100** to lock the lock **90** or maintains the status of the powered actuator **114** and/or the release lever **100** to keep the lock **90** locked.

[0021] A corresponding operation for a front crash can also be utilized. For example, in accordance with the track position sensor **134** indicating that the second track **62** is not in the second position and the crash sensor **132** indicates a front crash, the controller **130** operates the powered actuator **114** to actuate the release lever **100** to unlock the lock **90** and allows the second track **62** to move toward or to the second position, such as via the crash forces. Additionally or alternatively, in accordance with the track position sensor **134** indicating that the second track **62** is already in the second position with a front crash, the controller **130** actuates the powered actuator **114** to actuate the release lever **100** to lock the lock **90** or maintains the status of the powered actuator **114** and/or the release lever **100** to keep the lock **90** locked.

[0022] The second position can, for example, include a zero-gravity position or a slouched position. The first position can, for example, include a design or driving position in which the seat base **40** is closer to parallel with the mounting surface **26** and/or in which the seat back **42** is closer to perpendicular to the mounting surface **26**, which may be better suited for protecting the occupant **28** during certain crashes (e.g., rear crashes) than the second position. Additionally or alternatively, the second position may be better suited for protecting the occupant **28** during certain other crashes (e.g., front crashes) than the first position.

[0023] The track assembly **20** includes a first end stop **150** and a second end stop **152** that limit sliding movement of the second track **62** along the first track **60** (e.g., rotational movement of the seat assembly **24** about the rotational axis **80**). The first end stop **150** is disposed in the rear section **74** of the first track **60**, such as at or spaced from the rear end of the rear section **74**. The second end stop **152** is disposed in the front section **72** of the first track **60**, such as at or spaced from the front end of the front section **72**. The first end stop **150** limits movement of the second track **62** in a first rotational direction (e.g., counterclockwise in FIG. 2), which is generally rearward and upward relative to the first track **60**. The second end stop **152** limits movement of the second track **62** in a second rotational direction (e.g., clockwise in FIG. 2), which is generally forward and upward relative to the first track **60**. Optionally, the first end stop **150** is aligned with the first position of the second track **62** such that movement of the second track **62** (e.g., during a crash) is automatically stopped at the first position by contacting the first end stop **150**, such as in addition to or instead of the controller **130** operating the powered actuator **114** (e.g., if the track assembly **20** includes the mechanical actuator **112** and not the powered actuator **114**).

[0024] Referring to FIG. 3, the track assembly **20** is shown with a first damper **160** at the first end stop **150** and a second damper **162** at the second end stop **152**. The first damper **160** absorbs at least some of the force from the second track **62** as the second track **62** moves to the first end stop **150**, which can include crash forces. The second damper **162** absorbs at least some of the force from the second track **62** as the second track **62** moves to the second end stop **152**. While the first and second dampers **160**, **162** are shown at the first and second end stops **150**, **152**, the first damper **160** and/or the second damper **162** can be coupled to move with the second track **62** and engage the first end stop **150** and the second end stop **152**, respectively. The track assembly **20** is also shown with a track damper **164**. The track damper **164** is coupled to the first track **60** or the second track **62** to contact the other of the first track **60** or the second track **62** as the second track **62** moves along the first track **60**, such as to limit the speed of the second track **62** (e.g., via friction). The dampers **160**, **162**, **164** may limit forces applied to the occupant **28**.

[0025] Referring to FIG. 4, the track assembly **20** is shown with a third track **180** and a fourth track **182** that mount the first track **60** to the mounting surface **26**. The third track **180** is slidably connected to the fourth track **182**, which is mounted to the mounting surface **26**. The first track **60** is mounted to move with the third track **180** along the fourth track **182**. The third track **180** and the fourth track **182** have linear configurations that are parallel with the longitudinal direction X. For example, the third track **180** and the fourth track **182** provide longitudinal adjustment for the seat assembly **24** in addition to the rotational adjustment of the seat assembly **24** provided via the first track **60** and the second track **62** and/or the rotational adjustment of the seat back **42**, such as via the recliner **44**.

[0026] The track assembly **20** is shown with a frame **190** including a first frame portion **192** and a second frame portion **194**. The first frame portion **192** is coupled with the second track **62**. The second frame portion **194** is coupled with the seat assembly **24**. A pivot **196** rotatably couples the first frame portion **192** with the second frame portion **194** such that the seat assembly **24** is rotatable relative to the second track **62** about a seat axis **198**. The seat axis **198** extends in an X-Z plane and/or is perpendicular to the rotational axis **80**, which can be parallel with the lateral direction Y (e.g., the seat assembly **24** can be rotatable about two axes that are perpendicular to each other). Additionally or alternatively, the seat axis **198** is perpendicular to the second track **62**

and/or not parallel with the seat back axis **48**. When the second track **62** is in the first position, the seat axis **198** can be parallel or close to parallel with the vertical direction Z. The pivot **196** allows the seat assembly **24** to rotate relative to the second track **62**, such as between a forward-facing orientation shown in FIG. **4**, with the seat assembly **24** facing the longitudinal direction X, and one or more other orientations, such as the lateral-facing orientation shown in FIG. **5** in which the seat assembly **24** is facing the lateral direction Y. The rotational orientations of the seat assembly **24** relative to the second track **62** are not limited to facing the longitudinal and lateral directions X, Y, and can include positions between the longitudinal and lateral directions X, Y. The pivot **196** can be operated manually, can include or be coupled to a motor **200** for powered adjustment, or both. While the frame **190** and the pivot **196** are shown as rotatably coupling the seat assembly **24** with the second track **62**, the frame **190** and/or the pivot **196** can rotatably connect the first track **60** with the third track **180** or rotatably connect the fourth track **182** with the mounting surface **26**. In some configurations, the frame **190** can adjust the height of the seat assembly **24**. For example, one or both of the first frame portion **192** or the second frame portion **194** can include rotatable links **202**. [0027] The seat back **42** is rotatable about the seat back axis **48**, the rotational axis **80**, and the seat axis **198**. The seat back axis **48** and the rotational axis **80** are parallel and offset such that rotation of the seat back **42** about either or both axes modifies a recline angle **220** (FIG. **4**) of the seat back **42** relative to the mounting surface **26**. In some examples, the recline angle **220** of the seat back **42** is adjustable by at least 10 degrees and less than or equal to 90 degrees via movement about the rotational axis **80**.

[0028] Referring to FIG. **6**, the seat assembly **24** can rotate about the seat axis **198** and the rotational axis **80**. For example, the occupant **28** may rotate the seat assembly **24** about the seat axis **198** to reorient the seat assembly **24** to face a different part of the vehicle **30**, such as toward another occupant. In the event of a crash, the actuator **110** automatically unlocks the lock **90** and the crash forces cause the seat assembly **24** to rotate about the rotational axis **80** (e.g., counterclockwise in FIGS. **5** and **6**), such as from the position shown in FIG. **5** to the position shown in FIG. **6**. The rotation about the rotational axis **80** includes sliding movement of the second track **62** along the first track **60**.

[0029] Referring to FIG. **7**, a method **300** of operating a track assembly **20** includes unlocking the first and second tracks **60**, **62** (block **302**), and sliding the second track **62** along the first track **60** from an initial track position to a different track position, the first track **60** and the second track **62** having curved configurations (block **304**). The method **300** includes locking the second track **62** with the first track **60** at the second track position (block **306**), such as via the lock **90**. The method **300** includes sensing a crash (block **308**) and unlocking the second track **62** from the first track **60** in response to sensing the crash (block **310**). The crash **308** can be sensed via the crash sensor **132**. Additionally or alternatively, the mechanical actuator **112** may function as a crash sensor and perform both of blocks **308** and **310**. The method **300** includes sliding the second track **62** along the first track **60** to a crash position utilizing momentum from the crash (block **312**), and locking the second track **62** in the crash position (block **314**), which can include automatically locking the second track **62** at the crash track position. If the crash is a rear crash, the crash position can include the first track position in which the seat assembly **24** and the occupant **28** are more upright. If the crash is a front crash, the crash position can include the second track position in which the seat assembly **24** and the occupant **28** are closer to horizontal. For example, the controller **130** may dynamically determine the crash position as the first track position or the second track position according to whether the sensed crash is a rear crash or a front crash, respectively.

[0030] The first end stop **150** may automatically stop the second track **62** at the first position, such as to facilitate the locking in block **312**. Sliding the second track **62** to or toward the first track position can include the second track moving rearward and upward, or the second track **62** moving forward and upward, such as depending on the angle of impact. For example, with a rear crash, the second track **62** may slide rearward and upward toward the first position, and with a front crash, the

second track **62** may slide forward and upward toward the second position.

[0031] Automatically locking the second track **62** can be conducted via the mechanical actuator **112**, via the powered actuator **114**, or a combination thereof. For example, the controller **130** may actuate (e.g., release) the release lever **100** such that the release lever **100** causes or allows the lock **90** to return to a locked state when the second track **62** reaches the first track position. The seat back **42** and the head restraint **46** are more upright in the first track position than in the second track position.

[0032] The method **300** can, with some configurations, function as a method of limiting movement of a head **210** of the occupant **28**. For example, with the second track **62** in the first track position, the head restraint **46** of the seat assembly **24** is positioned to limit movement of the head **210** of the occupant **28** to a greater extent than with the second track **62** in the second track position. During a crash, the head restraint **46** can limit movement of the head **210** of the occupant **28** and the seat back **42** can limit movement of a torso **212** of the occupant **28**. In the second position of the second track **62**, the seat back **42** and/or the head restraint **46** may be closer to parallel with the mounting surface **26**, which may limit movement of the torso **212** and the head **210**, respectively, to a lesser extent than when the second track **62** is in the first position and the seat back **42** and the head restraint **46** are closer to perpendicular to the mounting surface **26**.

[0033] The method **300** can be carried out, at least in part, via the controller **130**. For example, the controller **130** may receive occupant input to move the second track **62** along the second track **62** from the first position to the second position, control the actuator **110** to unlock the tracks **60**, **62** (block **302**), control a track motor **240** (FIG. **1**) to cause the sliding movement, and control the actuator **110** to lock the tracks **60**, **62** at the second position. The controller **130** may determine the crash has occurred or is imminent via the crash sensor **132** (block **308**), unlock the tracks **60**, **62** (block **310**), such as to allow the sliding movement in block **312**, and then control the actuator **110** to lock the tracks **60**, **62** when the second track **62** reaches the first track position (block **314**).

[0034] With the method **300**, blocks **302-306** can be utilized for comfort adjustments of the seat assembly **24**, such as to improve the comfort of the occupant **28**, and blocks **310-314** can be utilized for crash adjustments of the seat assembly **24**, such as to improve the safety of the occupant **28** when a crash is sensed in block **308**.

[0035] The instant disclosure includes the following non-limiting embodiments:

[0036] A track assembly, comprising: a first track having a curved configuration; a second track slidably connected to the first track; a seat assembly connected to move with the second track, the seat assembly comprising a seat base and a seat back adjustable to a plurality of seat positions relative to the second track; a lock configured to selectively lock the first track and the second track; and a release lever configured to automatically actuate the lock.

[0037] The track assembly of any preceding embodiment, wherein an angular extent of the first track is at least 30 degrees and less than or equal to 120 degrees.

[0038] The track assembly of any preceding embodiment, wherein the angular extent of the first track is at least 40 degrees and less than or equal to 50 degrees.

[0039] The track assembly of any preceding embodiment, further comprising a mechanical actuator operably coupled with the release lever to automatically unlock the lock.

[0040] The track assembly of any preceding embodiment, further comprising an actuator operably coupled with the release lever to automatically unlock the lock, the actuator including at least one of a motor, an electromagnet, a piezoelectric material, or a pyrotechnic device.

[0041] The track assembly of any preceding embodiment, further comprising an electronic controller controlling operation of the actuator.

[0042] The track assembly of any preceding embodiment, further comprising a crash sensor connected to the electronic controller; wherein the electronic controller is configured to operate the actuator to actuate the release lever and disengage the lock in accordance with the crash sensor detecting a crash.

[0043] The track assembly of any preceding embodiment, wherein the seat assembly is rotatable about a first axis via the second track sliding along the first track, and is rotatable about a second axis via a pivot rotatably coupling the seat assembly with the second track; and the second axis is perpendicular to the second track and not parallel with a seat back axis about which the seat back is adjustable.

[0044] The track assembly of any preceding embodiment, further comprising a third track slidably connected to a fourth track, the fourth track having a linear configuration; wherein the first track is mounted to the third track.

[0045] The track assembly of any preceding embodiment, wherein the seat back is adjustable about a seat back axis and is adjustable about a rotational axis that is offset from the seat back axis such that a recline angle of the seat back is adjustable via movement of the seat back about the seat back axis and via movement of the seat assembly about the rotational axis.

[0046] The track assembly of any preceding embodiment, wherein an angular extent of the second track is less than the angular extent of the first track.

[0047] The track assembly of any preceding embodiment, wherein the angular extent of the second track is at least 25 degrees and less than or equal to 40 degrees.

[0048] The track assembly of any preceding embodiment, wherein the first track has a radius of at least 300 mm and less than or equal to 900 mm.

[0049] The track assembly of any preceding embodiment, further comprising an electronic controller controlling operation of the actuator.

[0050] The track assembly of any preceding embodiment, further comprising a crash sensor connected to the electronic controller; wherein the electronic controller is configured to operate the actuator to actuate the release lever and disengage the lock in accordance with the crash sensor detecting a crash.

[0051] The track assembly of any preceding embodiment, wherein the electronic controller is configured to operate the actuator to reengage the lock when the second track reaches a crash position.

[0052] The track assembly of any preceding embodiment, further comprising a track position sensor sensing a position of the second track relative to the first track.

[0053] The track assembly of any preceding embodiment, wherein the electronic controller is configured to operate the actuator to reengage the lock according, at least in part, to the position of the second track obtained via the track position sensor.

[0054] The track assembly of any preceding embodiment, wherein the first track includes a first end stop and a second end stop that limit rotational movement of the seat assembly.

[0055] The track assembly of any preceding embodiment, wherein the first track comprises a damper disposed at the first end stop or the second end stop.

[0056] The track assembly of any preceding embodiment, wherein the second track includes a damper configured to engage the first end stop or the second end stop.

[0057] The track assembly of any preceding embodiment, wherein the seat assembly is rotatable about a first axis via the second track sliding along the first track, and is rotatable about a second axis via a pivot rotatably coupling the seat assembly with the second track.

[0058] The track assembly of any preceding embodiment, further comprising a third track slidably connected to a fourth track, the fourth track having a linear configuration; wherein the first track is mounted to the third track.

[0059] The track assembly of any preceding embodiment, wherein the seat back is adjustable about a seat back axis and is adjustable about a rotational axis that is offset from the seat assembly axis such that a recline angle of the seat back is adjustable via movement of the seat back about the seat back axis and via movement of the seat back about the rotational axis.

[0060] The track assembly of any preceding embodiment, wherein the angle of the seat back is adjustable by at least 10 degrees and less than or equal to 90 degrees via movement of the seat back

about the track assembly axis.

[0061] A vehicle, comprising: a mounting surface; and the track assembly of any preceding embodiment, coupled to the mounting surface.

[0062] The vehicle of any preceding embodiment, wherein the seat assembly is rotatable about two axes that are perpendicular to each other.

[0063] The vehicle of any preceding embodiment, wherein the track assembly comprises a third track slidably connected to a fourth track that is mounted to the mounting surface and has a linear configuration; and the first track is mounted to third track.

[0064] A method of operating a track assembly, the method comprising: sliding a second track along a first track to a first track position, the first track having a curved configuration; locking the second track with the first track at the first track position; sensing a crash; unlocking the second track from the first track in response to sensing the crash; sliding the second track along the first track to a second track position utilizing momentum from the crash; and locking the second track in the second track position.

[0065] The method of any preceding embodiment, wherein the first track includes a first end stop and a second end stop; and the second track position is at the first end stop or the second end stop.

[0066] The method of any preceding embodiment, wherein a seat assembly is connected to the second track and includes a seat base and a seat back; and the seat back is more upright in the second track position than the first track position.

[0067] The method of any preceding embodiment, further comprising damping, via a damper, the sliding of the second track to the second track position.

[0068] The method of any preceding embodiment, wherein the damper is disposed at the first end stop or the second end stop.

[0069] The method of any preceding embodiment, further comprising, after sliding the second track to the second track position, limiting movement of a head of an occupant of the seat assembly via a head restraint of the seat assembly.

[0070] The method of any preceding embodiment, wherein sliding the second track to the second track position includes the second track moving rearward and upward.

[0071] The method of any preceding embodiment, wherein sliding the second track to the second track position includes the second track moving forward and upward.

[0072] The method of any preceding embodiment, comprising, prior to sensing the crash and sliding the second track to the second track position, rotating a seat coupled to the second track from a forward orientation to a transverse orientation.

[0073] A method of limiting movement of a head of an occupant of a vehicle seat assembly during a crash, the method comprising: sliding a second track along a first track from an initial track position to a different track position, the first track having a curved configuration; locking the second track with the first track at the different track position; sensing the crash; automatically unlocking the second track from the first track in response to sensing the crash; sliding the second track along the first track to a crash position utilizing momentum from the crash; and locking the second track in the crash position.

[0074] The method of any preceding embodiment, wherein, in the first track position, a head restraint of the vehicle seat assembly is positioned to limit movement of the head of the occupant to a greater extent than with the second track in the second track position.

[0075] A method of limiting movement of a head of an occupant of a vehicle seat assembly during a crash, the method comprising: sliding a second track along a first track from an initial track position to a different track position, the first track having a curved configuration; locking the second track with the first track at the different track position; sensing the crash; automatically unlocking the second track from the first track in response to sensing the crash; sliding the second track along the first track from the track position to a crash position utilizing momentum from the crash; and locking the second track in the crash position; wherein, in the crash position, a head

restraint of the vehicle seat assembly is positioned to limit movement of the head of the occupant to a greater extent than with the second track in the track position.

[0076] An electronic controller configured to implement the method of any preceding embodiment.

[0077] A non-transitory computer-readable storage medium having a computer program encoded thereon for implementing the method of any preceding embodiment.

[0078] In examples, a controller (e.g., the electronic controller **130**) may include an electronic controller and/or include an electronic processor, such as a programmable microprocessor and/or microcontroller. In embodiments, a controller may include, for example, an application specific integrated circuit (ASIC) and/or an embedded controller. A controller may include a central processing unit (CPU), a memory (e.g., a non-transitory computer-readable storage medium), and/or an input/output (I/O) interface. A controller may be configured to perform various functions, including those described in greater detail herein, with appropriate programming instructions and/or code embodied in software, hardware, and/or other medium. In embodiments, a controller may include a plurality of controllers. In embodiments, a controller may be connected to a display, such as a touchscreen display.

[0079] Various examples/embodiments are described herein for various apparatuses, systems, and/or methods. Numerous specific details are set forth to provide a thorough understanding of the overall structure, function, manufacture, and use of the examples/embodiments as described in the specification and illustrated in the accompanying drawings. It will be understood by those skilled in the art, however, that the examples/embodiments may be practiced without such specific details. In other instances, well-known operations, components, and elements have not been described in detail so as not to obscure the examples/embodiments described in the specification. Those of ordinary skill in the art will understand that the examples/embodiments described and illustrated herein are non-limiting examples, and thus it can be appreciated that the specific structural and functional details disclosed herein may be representative and do not necessarily limit the scope of the embodiments.

[0080] Reference throughout the specification to “examples,” “in examples,” “with examples,” “various embodiments,” “with embodiments,” “in embodiments,” “an embodiment,” “with some configurations,” “in some configurations,” or the like, means that a particular feature, structure, or characteristic described in connection with the example/embodiment is included in at least one embodiment. Thus, appearances of the phrases “examples,” “in examples,” “with examples,” “in various embodiments,” “with embodiments,” “in embodiments,” “an embodiment,” “with some configurations,” “in some configurations,” or the like, in places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, and/or characteristics may be combined in any suitable manner in one or more examples/embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment/example may be combined, in whole or in part, with the features, structures, functions, and/or characteristics of one or more other embodiments/examples without limitation given that such combination is not illogical or non-functional. Moreover, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the scope thereof. The word “exemplary” is used herein to mean “serving as a non-limiting example.”

[0081] It should be understood that references to a single element are not necessarily so limited and may include one or more of such element, unless the context clearly indicates otherwise. Any directional references (e.g., plus, minus, upper, lower, upward, downward, left, right, leftward, rightward, top, bottom, above, below, vertical, horizontal, clockwise, and counterclockwise) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of examples/embodiments.

[0082] “One or more” includes a function being performed by one element, a function being

performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above. The term “at least one of” in the context of, e.g., “at least one of A, B, and C” or “at least one of A, B, or C” includes only A, only B, only C, or any combination or subset of A, B, and C, including any combination or subset of one or a plurality of A, one or a plurality of B, and one or a plurality of C.

[0083] Although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the various described embodiments. The first element and the second element are both elements, but they are not the same element.

[0084] The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. Uses of “and” and “or” are to be construed broadly (e.g., to be treated as “and/or”). For example and without limitation, uses of “and” do not necessarily require all elements or features listed, and uses of “or” are inclusive unless such a construction would be illogical. The terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0085] Joinder references (e.g., attached, coupled, connected, and the like) are to be construed broadly and may include intermediate members between a connection of elements, relative movement between elements, direct connections, indirect connections, fixed connections, movable connections, operative connections, indirect contact, and/or direct contact. As such, joinder references do not necessarily imply that two elements are directly connected/coupled and in fixed relation to each other. Connections of electrical components, if any, may include mechanical connections, electrical connections, wired connections, and/or wireless connections, among others. Uses of “e.g.” and “such as” in the specification are to be construed broadly and are used to provide non-limiting examples of embodiments of the disclosure, and the disclosure is not limited to such examples.

[0086] While processes, systems, and methods may be described herein in connection with one or more steps in a particular sequence, such methods may be practiced with the steps in a different order, with certain steps performed simultaneously, with additional steps, and/or with certain described steps omitted.

[0087] As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

[0088] All matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative only and not limiting. Changes in detail or structure may be made without departing from the present disclosure.

[0089] A controller, an electronic control unit (ECU), a system, and/or a processor as described herein may include a conventional processing apparatus known in the art, which may be capable of executing preprogrammed instructions stored in an associated memory, all performing in

accordance with the functionality described herein. To the extent that the methods described herein are embodied in software, the resulting software can be stored in an associated memory and can also constitute means for performing such methods. Such a system or processor may further be of the type having ROM, RAM, RAM and ROM, and/or a combination of non-volatile and volatile memory so that any software may be stored and yet allow storage and processing of dynamically produced data and/or signals.

[0090] An article of manufacture in accordance with this disclosure may include a non-transitory computer-readable storage medium having a computer program encoded thereon for implementing logic and other functionality described herein. The computer program may include code to perform one or more of the methods disclosed herein. Such embodiments may be configured to execute via one or more processors, such as multiple processors that are integrated into a single system or are distributed over and connected together through a communications network, and the communications network may be wired and/or wireless. Code for implementing one or more of the features described in connection with one or more embodiments may, when executed by a processor, cause a plurality of transistors to change from a first state to a second state. A specific pattern of change (e.g., which transistors change state and which transistors do not), may be dictated, at least partially, by the logic and/or code.

Claims

1. A track assembly, comprising: a first track having a curved configuration; a second track slidably connected to the first track; a seat assembly connected to move with the second track, the seat assembly comprising a seat base and a seat back adjustable to a plurality of seat positions relative to the second track; a lock configured to selectively lock the first track and the second track; and a release lever configured to automatically actuate the lock.
2. The track assembly of claim 1, wherein an angular extent of the first track is at least 30 degrees and less than or equal to 120 degrees.
3. The track assembly of claim 2, wherein the angular extent of the first track is at least 40 degrees and less than or equal to 50 degrees.
4. The track assembly of claim 1, further comprising a mechanical actuator operably coupled with the release lever to automatically unlock the lock.
5. The track assembly of claim 1, further comprising an actuator operably coupled with the release lever to automatically unlock the lock, the actuator including at least one of a motor, an electromagnet, a piezoelectric material, or a pyrotechnic device.
6. The track assembly of claim 5, further comprising an electronic controller controlling operation of the actuator.
7. The track assembly of claim 6, further comprising a crash sensor connected to the electronic controller; wherein the electronic controller is configured to operate the actuator to actuate the release lever and disengage the lock in accordance with the crash sensor detecting a crash.
8. The track assembly of claim 1, wherein the seat assembly is rotatable about a first axis via the second track sliding along the first track, and is rotatable about a second axis via a pivot rotatably coupling the seat assembly with the second track; and the second axis is perpendicular to the second track and not parallel with a seat back axis about which the seat back is adjustable.
9. The track assembly of claim 8, further comprising a third track slidably connected to a fourth track, the fourth track having a linear configuration; wherein the first track is mounted to the third track.
10. The track assembly of claim 1, wherein the seat back is adjustable about a seat back axis and is adjustable about a rotational axis that is offset from the seat back axis such that a recline angle of

the seat back is adjustable via movement of the seat back about the seat back axis and via movement of the seat assembly about the rotational axis.
